

Metallurgy

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Q1 JEE Main 2020 - 6 September (Morning)

The incorrect statement is

- (A) bronze is an alloy of copper and tin
- (B) cast iron is used to manufacture wrought iron
- (C) german silver is an alloy of zinc, copper and nickel
- (D) brass is an alloy of copper and nickel

Q2 JEE Main 2020 - 2 September (Evening)

Cast iron is used for the manufacture of

- (A) wrought iron, pig iron and steel
- (B) pig iron, scrap iron and steel
- (C) wrought iron and pig iron
- (D) wrought iron and steel

Q3 JEE Main 2020 - 4 September (Morning)

Among statements (a) - (d), the correct ones are

- (a) lime stone is decomposed to CaO during the extraction of iron from its oxides.
- (b) In the extraction of silver, silver is extracted as an anionic complex.
- (c) Nickel is purified by Mond's process.
- (d) Zr and Ti are purified by Van Arkel method.

- (A) (a), (c) and (d) only
- (B) (c) and (d) only
- (C) (b), (c) and (d) only
- (D) (a), (b), (c) and (d)

Q4 JEE Main 2020 - 4 September (Evening)

The processes of calcination and roasting in metallurgical industries, respectively, can lead to

- (A) Global warming and photochemical smog
- (B) Photochemical smog and ozone layer depletion
- (C) Photochemical smog and global warming
- (D) Global warming and acid rain

Metallurgy

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Q5 JEE Main 2020 - 5 September (Morning)

An Ellingham diagram provides information about

- (A) the temperature dependence of the standard Gibbs energies of formation of some metal oxides
- (B) the pressure dependence of the standard electrode potentials of reduction reactions involved in the extraction of metals
- (C) the conditions of pH and potential under which a species is thermodynamically stable
- (D) the kinetics of the reduction process

Q6 JEE Main 2020 - 5 September (Evening)

Boron and silicon of very high purity can be obtained through

- (A) vapour phase refining
- (B) electrolytic refining
- (C) liquation
- (D) zone refining

Q7 JEE Main 2020 - 6 September (Evening)

The element that can be refined by distillation is

- (A) gallium
- (B) nickel
- (C) zinc
- (D) tin

Q8 JEE Main 2020 - 7 January (Morning)

Purest form of commercial iron is:

- (A) Wrought iron
- (B) Cast iron
- (C) Pig iron
- (D) Iron scrap + pig iron

Q9 JEE Main 2020 - 7 January (Evening)

Metal with low melting point containing impurities of high melting point can be purified by

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Metallurgy

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

(A) Zone refining

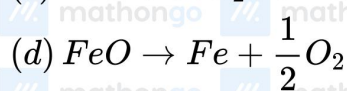
(B) Vapor phase refining

(C) Distillation

(D) Liquation

Q10 JEE Main 2020 - 8 January (Evening)

Among the reactions (a) - (d), the reaction(s) that does/do not occur in the blast furnace during the extraction of iron is/are



(A) a & b

(B) a, b & c

(C) c & d

(D) a, b, c, d

Metallurgy

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Answer Key

Q1 (D)

Q2 (D)

Q3 (D)

Q4 (D)

Q5 (A)

Q6 (D)

Q7 (C)

Q8 (A)

Q9 (D)

Q10 (C)

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Metallurgy

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Q1

Bronze contains 88 – 96% Cu and 4 – 12% Sn Cast iron is used to make wrought iron

German silver contains 50% Cu, 30% Zn and 20% Ni

Brass contains 70% Cu and 30% Zn.

Q2

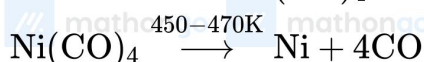
Cast iron is used for the manufacture of wrought iron and steel.

Q3



In extraction of silver, silver is extracted as an anionic complex $[\text{Ag}(\text{CN})_2]^-$

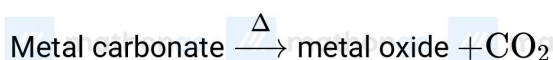
Ni is purified by Mond's process



Zr and Ti are purified by Van Arkel method.

Q4

Calcination : Heating of concentrated ore in the absence (or limited supply of air). Mainly used for carbonate ores.



(causes Global warming) Roasting : Heating of concentrated ore in a regular supply of air. Mainly used for sulphide ores.



(causes acid rain)

Q5

An Ellingham diagram provides information about, the temperature dependence of the standard gibbs energies of formation of some metal oxides.

Metallurgy

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Q6

For boron and silicon zone refining is used

Q7

zinc is refined by distillation

Q8

Purest form is wrought iron.

Q9

Theory Based

Q10

Theory Based

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